

BANKING DASHBOARD

Technical Documentation & Stakeholder Report

Author	Platform	Domain
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1. Executive Summary

Project Overview

This document presents the technical architecture and business logic behind a Banking Dashboard developed using Microsoft Power BI. The dashboard enables financial institutions to make data-driven lending decisions by analysing client profiles, loan portfolios, deposit patterns, and processing fees.

1.1 Problem Statement

Financial institutions face significant risk when extending credit to clients. Without robust data analytics, banks are exposed to default risk, poor capital allocation, and customer attrition. The core challenge is to develop a scalable risk analytics framework that uses historical and real-time client data to minimise financial exposure while maximising responsible lending.

1.2 Proposed Solution

The solution leverages Power BI's advanced data modelling and DAX (Data Analysis Expressions) capabilities to build four interactive dashboards. These dashboards aggregate multi-table banking data and surface actionable KPIs that help relationship managers and risk officers evaluate applicant profiles and make informed approval decisions.

2. Dataset Architecture

2.1 Dataset Description

The dataset models a relational banking environment with five interconnected tables linked via primary and foreign keys. This star-schema design ensures efficient querying and accurate aggregations across client segments.

Table Name	Description
Banking Relationship	Maps clients to their banking product relationships and advisor assignments.
Client-Banking	Core transactional table: deposits, loans, fees, engagement data per client.
Gender	Demographic dimension for gender-based segmentation and reporting.
Investment Advisor	Details of assigned advisors for each client relationship.

Table Name	Description
Period	Time dimension enabling trend analysis across reporting periods.

3. Data Engineering & Transformation

3.1 Derived Columns Created

The following calculated columns were added to the Client-Banking table to enrich the dataset:

- Engagement Timeframe — Categorises clients by their tenure phase with the bank (e.g. new, growing, loyal).
- Engagement Days — Calculates the total number of days since the client joined the bank using DATEDIFF.
- Income Band — Segments clients into Low (< ₹1,00,000), Mid (< ₹3,00,000), and High income brackets.
- Processing Fees — Assigns a fee rate based on the existing Fee Structure column (e.g. High = 5%).

4. DAX Functions Reference

The following Power BI DAX functions form the computational backbone of the dashboards:

SUM

Adds all numeric values in a specified column, returning a decimal result. Used for aggregating financial amounts across the portfolio.

Syntax:

SUM(<column>)

Example:

Bank Deposit = SUM('Clients - Banking'[Bank Deposits])

DISTINCTCOUNT

Counts the number of unique values in a column. Essential for calculating Total Clients without double-counting shared accounts.

Syntax:

```
DISTINCTCOUNT(<column>)
```

Example:

```
Total Clients = DISTINCTCOUNT('Clients - Banking'[Client ID])
```

SUMX

Iterates row-by-row over a table, evaluates an expression for each row, and returns the sum. Used to calculate fee amounts that depend on row-level loan values.

Syntax:

```
SUMX(<table>, <expression>)
```

Example:

```
Total Fees = SUMX('Clients - Banking', [Total Loan] * 'Clients - Banking'[Processing Fees])
```

SWITCH

Evaluates an expression against a list of values and returns the matching result. Used for conditional logic such as fee categorisation.

Syntax:

```
SWITCH(<expression>, <value>, <result>[, <value>, <result>]...[, <else>])
```

Example:

```
Income Band = SWITCH(TRUE(),  
  'Clients - Banking'[Estimated Income] < 100000, "Low",  
  'Clients - Banking'[Estimated Income] < 300000, "Mid",  
  "High")
```

DATEDIFF

Returns the number of interval boundaries between two dates. Used to calculate client engagement duration in days.

Syntax:

```
DATEDIFF(<Date1>, <Date2>, <Interval>)
```

Example:

```
Engagement Days = DATEDIFF('Clients - Banking'[Joined Bank], TODAY(), DAY)
```

5. KPI Catalogue

The table below enumerates all Key Performance Indicators, their business definitions, and their DAX formulas:

KPI Name	Description	DAX Formula
Total Clients	Unique count of all client IDs in the banking portfolio.	<code>DISTINCTCOUNT('Clients - Banking'[Client ID])</code>
Total Loan	Combined outstanding obligation: Bank Loan + Business Lending + Credit Cards Balance.	<code>[Bank Loan] + [Business Lending] + [Credit Cards Balance]</code>
Bank Loan	Sum of all bank loan amounts to be repaid by clients.	<code>SUM('Clients - Banking'[Bank Loans])</code>
Business Lending	Aggregate loan disbursements to small and medium enterprises.	<code>SUM('Clients - Banking'[Business Lending])</code>
Credit Cards Balance	Total outstanding credit card balances owed to the bank.	<code>SUM('Clients - Banking'[Credit Card Balance])</code>
Total Deposit	Sum of all deposit types: Bank + Savings + Foreign Currency + Checking.	<code>[Bank Deposit]+[Savings Account]+[Foreign Currency Account]+[Checking Accounts]</code>
Bank Deposit	Total money deposited directly into bank accounts.	<code>SUM('Clients - Banking'[Bank Deposits])</code>
Savings Account	Interest-bearing deposit balances held by clients.	<code>SUM('Clients - Banking'[Saving Accounts])</code>
Checking Accounts	Transaction account balances used for daily banking needs.	<code>SUM('Clients - Banking'[Checking Accounts])</code>
Foreign Currency	Balances held in non-domestic currencies (excl. INR, BTN, NPR).	<code>SUM('Clients - Banking'[Foreign Currency Account])</code>
Total CC Amount	Aggregate amount across all credit card products.	<code>SUM('Clients - Banking'[Amount of Credit Cards])</code>
Total Fees	Revenue earned by the bank via processing fee charges on loan amounts.	<code>SUMX('Clients - Banking', [Total Loan] * 'Clients - Banking'[Processing Fees])</code>

KPI Name	Description	DAX Formula
Engagement Length	Total client engagement days aggregated across the portfolio.	<code>SUM('Clients - Banking'[Engagement Days])</code>

KPI'S:

In which followings KPIS are present :

Total Clients :

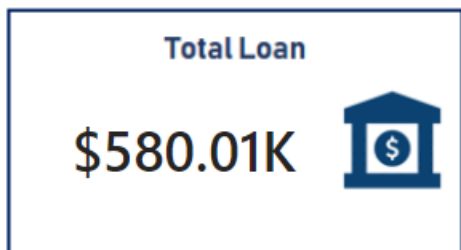
Total Clients KPI represents total number of clients in banking.

Total Clients = `DISTINCTCOUNT('Clients - Banking'[Client ID])`

**Total Loan :**

Total Loan gives you information about the bank loan + Business lending + credit cards balance of particular investor , gender.

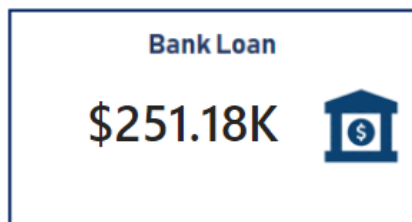
Total Loan = `[Bank Loan] + [Business Lending] + [Credit Cards Balance]`



Bank Loan :

Bank Loan gives you information what is the loan amount of loan to be repaid by the client to bank.

Bank Loan = SUM('Clients - Banking'[Bank Loans])

**Business Lending :**

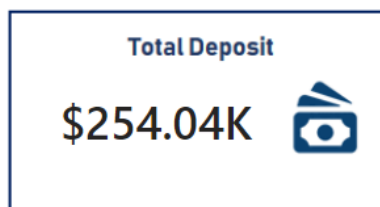
Business lending gives you information about the loan amount given to small business.

Business Lending = SUM('Clients - Banking'[Business Lending])

**Total Deposit**

Total Deposit gives you information about the amount deposited by particular investors in bank

Total Deposit = [Bank Deposit] + [Savings Account] + [Foreign Currency Account] + [Checking Accounts]

**Total Fees :**

Total Fees is nothing but the amount charged by the bank for account set-up , maintenance charges etc.

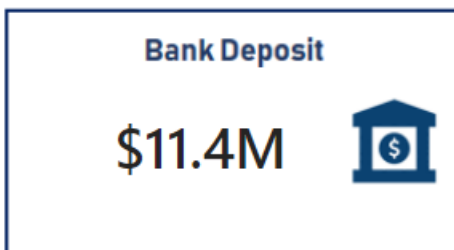
Total Fees = SUMX('Clients - Banking' , [Total Loan] * 'Clients - Banking'[Processing Fees])

**Bank Deposit :**

Bank deposit is the money put in the bank.

Bank Deposit =

$\text{SUM}(\text{'Clients - Banking'}[\text{Bank Deposits}])$

**Checking Account Amount :**

Checking account amount is nothing but which offers easy access to your money for daily transactional needs.

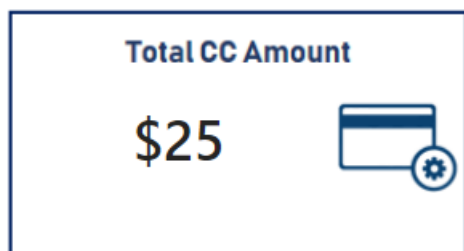
Checking Accounts =

$\text{SUM}(\text{'Clients - Banking'}[\text{Checking Accounts}])$

**Total CC Amount :**

Total CC Amount is a short-term source of financing for a company by a bank.

Total CC Amount = $\text{SUM}(\text{'Clients - Banking'}[\text{Amount of Credit Cards}])$

**Saving Account Amount :**

A savings account is an interest-bearing deposit account held at a bank.

Savings Account = SUM('Clients - Banking'[Saving Accounts])

**Foreign Currency Amount :**

Foreign Currency Account means an account held in a currency that is not the currency of India or Bhutan or Nepal.

Foreign Currency Account =
SUM('Clients - Banking'[Foreign Currency Account])

**Engagement Account :**

Engagement Banking is nothing but puts the customer at the center and aims to deliver the digital experiences they expect.

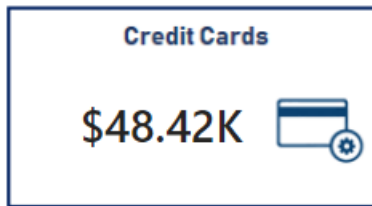
Engagement Length =
SUM('Clients - Banking'[Engagment Days])



Credit Cards Balance :

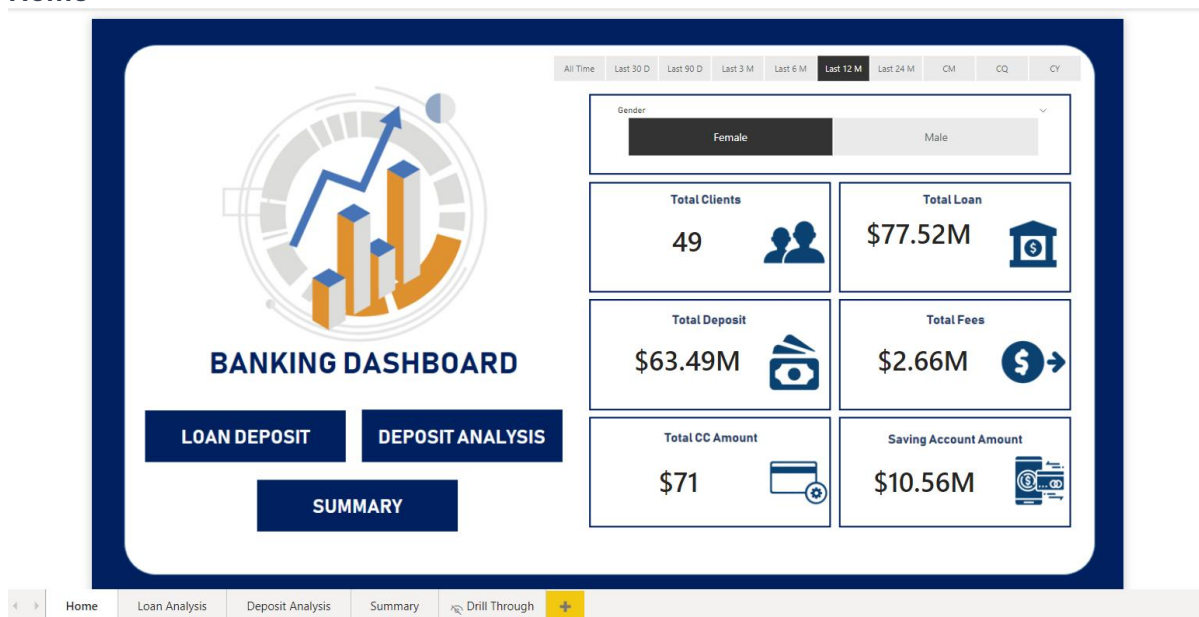
It is the total amount of money currently owned by a cardholder to their credit card bank.

Credit Cards Balance = SUM('Clients - Banking'[Credit Card Balance])

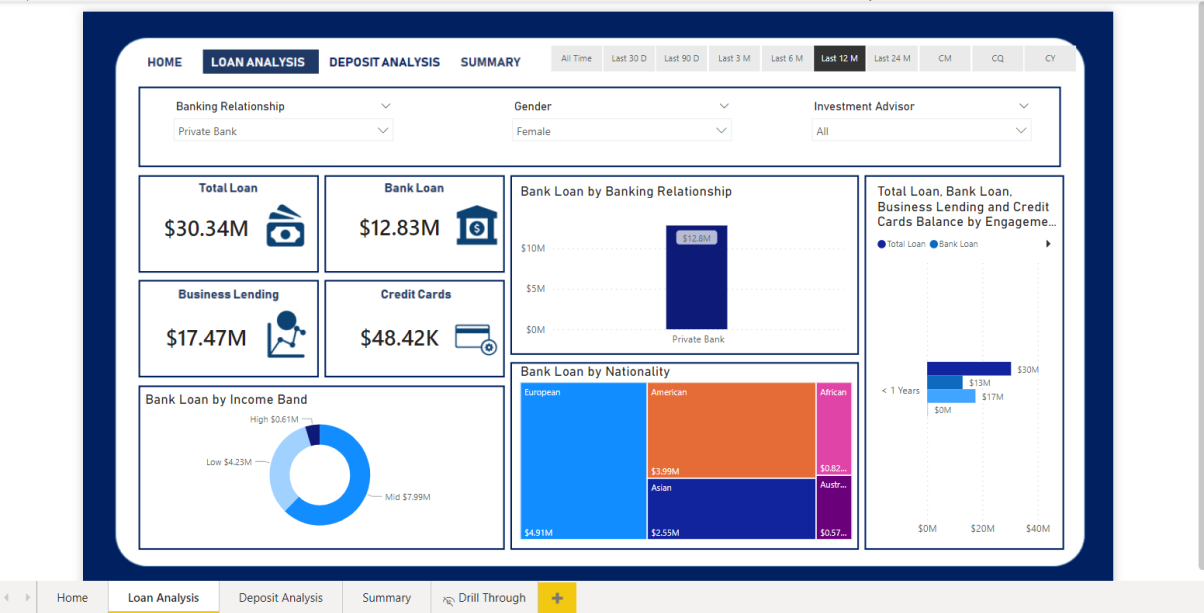


6. Dashboard Architecture

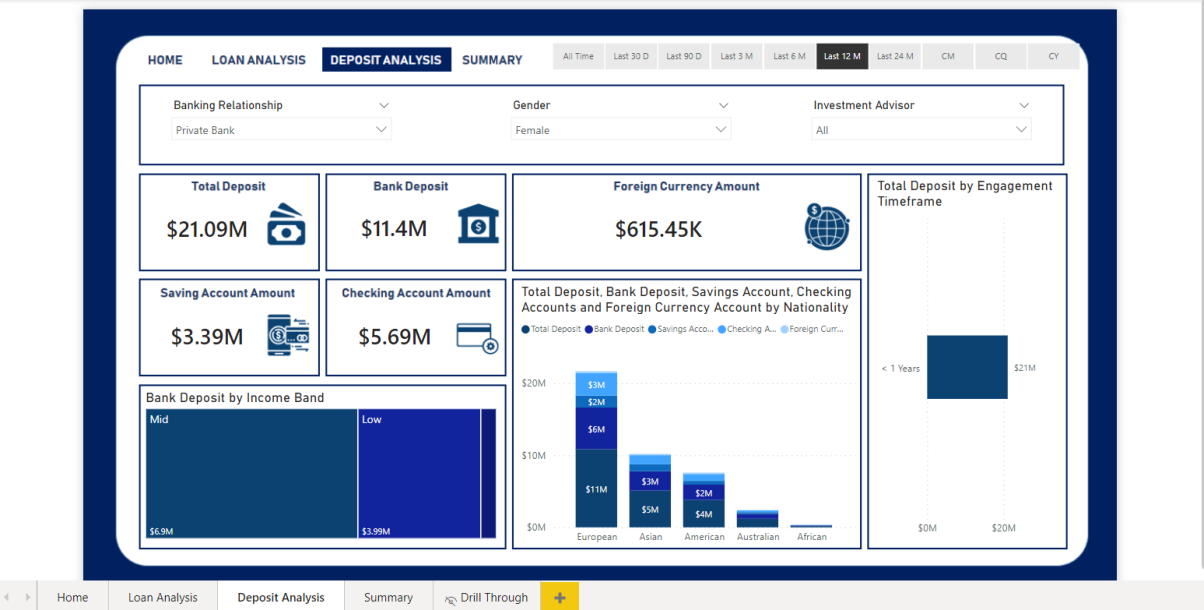
The solution consists of four purpose-built Power BI dashboards, each targeting a specific analytical need:

Visualization And Result –**Home**

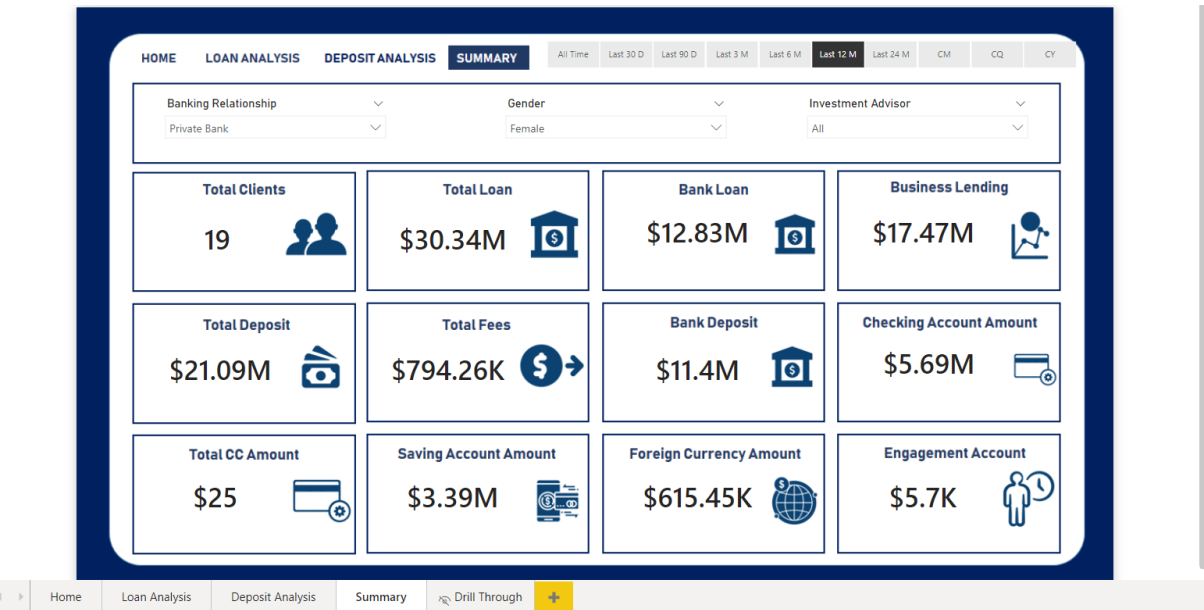
Loan Analysis



Deposit Analysis



Summary Dashboard



7. Key Business Insights

- Private banks attract significantly more clients than public-sector counterparts — a strategic benchmark for competitor analysis.
- Nationality-level drill-downs reveal which demographics carry the highest loan exposure, informing targeted risk mitigation.
- Deposit-to-loan ratios surfaced per investor type enable smarter capital allocation and credit scoring decisions.
- Processing fee modelling directly links the bank's fee revenue to total loan volume — enabling scenario planning.
- Engagement duration metrics help identify at-risk clients with declining activity before they churn.

8. Conclusion & Future Roadmap

8.1 Conclusion

This Banking Dashboard demonstrates how modern data visualisation with Power BI can transform raw relational banking data into actionable intelligence. By combining a well-structured data model with precise DAX measures and intuitive dashboards, the solution empowers banking teams to make faster, evidence-based decisions on lending, risk, and customer engagement.

8.2 Future Roadmap

- Integrate machine learning models (e.g. credit scoring classifiers) directly into the Power BI workflow via Azure ML.
- Add real-time data refresh pipelines connected to core banking systems for live dashboard updates.
- Expand geographic visualisations with map-based heat maps for branch performance benchmarking.
- Incorporate a customer lifetime value (CLV) metric to prioritise high-value client retention efforts.
- Build automated alert notifications when KPIs breach defined risk thresholds.

9. Author & Contact

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